

Graph 1 has the least gradient and thus highest c .

19. Answer: A

The pressure in the two glass bulbs must always be equal, since they are connected by the fine tube. The number of particles must also always stay constant as this is a closed system.

Prior to heating, the value of n can be represented using the ideal gas equation,

$$n = \frac{p_i V}{RT} = \frac{(1.00 \times 10^5)(0.60)}{300R}$$

After heating,

$$n = \frac{p_f V_1}{RT_1} + \frac{p_f V_2}{RT_2} = \frac{p_f (0.40)}{300R} + \frac{p_f (0.20)}{600R}$$

Solving for n , $n = 1.2 \times 10^5$ Pa

20. Answer: D